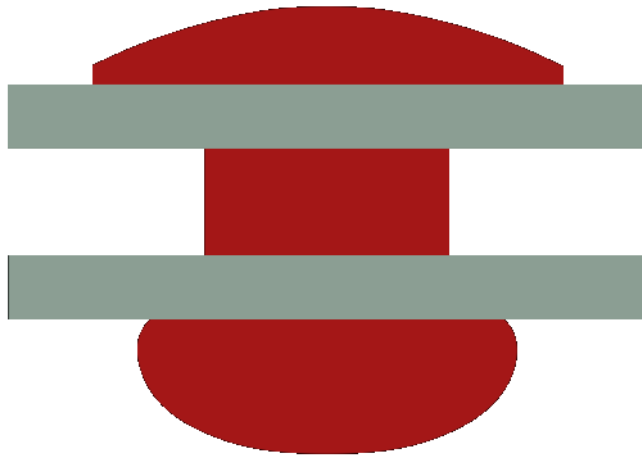




UDL Beam

Example 1 - rivt Doc

Attn: User Example

proj. 0001



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01.1 Summary

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam. It also serves as an annotated example of a rivt doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the [rivt user manual](#).

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command at the end of each rivt file (Doc-API *rv.D*). Published files are found in the *_published* folder.

01.2 Load Combinations

Table 1: ASCE 7-05 Load Effects (stored: t001-1.csv)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

01.3 Loads and Geometry

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 2: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-05 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 lb_in2	137.90 MPA	allowable stress

The VALTABLE command reads variable values from a file in the rvsr folder. The description is used as the table title. The range specifies the starting and ending line to be read from the file (0:0 means all lines).

Table 3: Beam Geometry (rvsrc/beam1.csv)

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing



variable	value	[value]	description
spn_1	16.00 ft	4.88 m	beam span

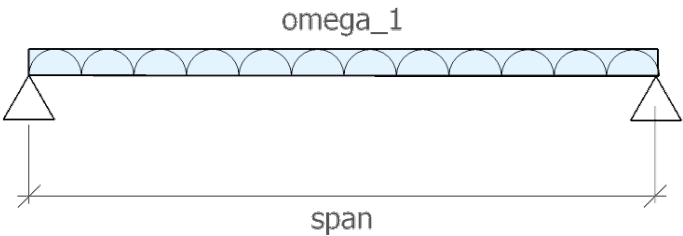


Fig. 1 - Beam Diagram

Uniform Distributed Loads

Eq. 1: Dead load [ASCE7-05 2.3.2]

$$dl_1 = 1.2 \cdot (D_4 + spc_1 \cdot (D_1 + D_2 + D_3))$$
$$dl_1 = 3.64 \text{ klf} \quad [dl_1] = 53.09 \text{ kN}_m \quad | \text{ Dead load [ASCE7-05 2.3.2]}$$

spc ₁	D ₄	D ₁	D ₃	D ₂
2.00 ft	3.00 klf	3.80 psf	10.00 psf	2.10 psf
beam spacing	fixed machinery DL	joists DL	partitions DL	plywood DL

Eq. 2: Live load [ASCE7-05 2.3.2]

$$ll_1 = 1.6 \cdot L_1 \cdot spc_1$$
$$ll_1 = 0.13 \text{ klf} \quad [ll_1] = 1.87 \text{ kN}_m \quad | \text{ Live load [ASCE7-05 2.3.2]}$$

spc ₁	L ₁
2.00 ft	40.00 psf
beam spacing	ASCE7-05 LL

Eq. 3: Total load [ASCE7-05 2.3.2]

$$w_1 = dl_1 + ll_1$$

$$w_1 = 3.77 \text{ klf} \quad [w_1] = 54.96 \text{ kN}_m \quad | \text{ Total load [ASCE7-05 2.3.2]}$$

ll_1	dl_1
128.00 ft·psf	3.64 klf
Live load [ASCE7-05 2.3.2]	Dead load [ASCE7-05 2.3.2]

01.4 Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4: Beam functions (rvsrc/sectprop.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4: rectangle - S (sectprop.py)

$$section_1 = rectsect(b_1, h_1)$$

$$section_1 = 540.00 \text{ in}^3 \quad [section_1] = 8849.01 \text{ cm}^3 \quad | \text{ rectangle - S (sectprop.py)}$$

h_1	b_1
18.00 inch	10.00 inch
beam depth	beam width

Eq. 5: rectangle - I (sectprop.py)

$$inertia_1 = rectinertia(b_1, h_1)$$

$$inertia_1 = 4860.0 \text{ in}^4 \quad [inertia_1] = 202288.5 \text{ cm}^4 \quad | \text{ rectangle - I (sectprop.py)}$$

h_1	b_1
18.0 inch	10.0 inch



beam depth beam width

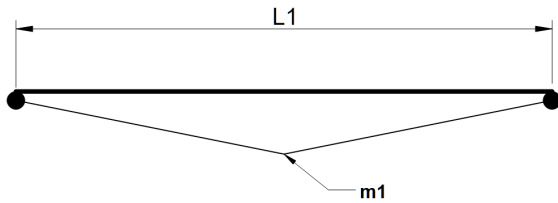


Fig. 2 - Moment diagram

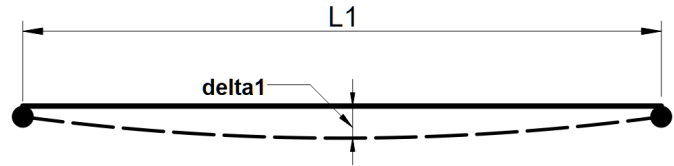


Fig. 3 - Deflection diagram

Eq.6: Maximum bending stress formula

$$\sigma_1 = \frac{M_1}{S_1}$$

Eq. 7: Mid-span UDL moment

$$m_1 = \frac{w_1 \cdot spn_1^2}{8}$$

$$m_1 = 120.52 \text{ ftkip} \quad [m_1] = 163.40 \text{ mkN} \quad | \text{ Mid-span UDL moment}$$

w_1	spn_1
3.77 klf	16.00 ft
Total load [ASCE7-05 2.3.2]	beam span

Eq. 8: Bending stress

$$fb_1 = \frac{m_1}{\text{section}_1}$$

$$fb_1 = 2678.2 \text{ lb_in}^2 \quad [fb_1] = 18.5 \text{ MPA} \quad | \text{ Bending stress}$$



<u>section₁</u>	<u>m₁</u>
540.0 inch ³	120.5 ft ² ·klf
rectangle - S (sectprop.py)	Mid-span UDL moment

Eq. 9: Stress ratio

fb ₁ < Fb ₁				
<u>fb₁</u>	<u>Fb₁</u>	<u>ratio L/R</u>	<u>check</u>	<u>reference</u>
2.68 ksi	20.00 ksi	0.133908	OK	Stress ratio

Eq. 10: mid-span deflection (sectprop.py)

δ ₁ = midspan_δ(spn ₁ , ω ₁ , E ₁ , inertia ₁)			
δ ₁ = 0.04 inch [δ ₁] = 1.00 mm mid-span deflection (sectprop.py)			
<u>E₁</u>	<u>ω₁</u>	<u>inertia₁</u>	<u>spn₁</u>
29000.00 ksi	3.77 klf	4860.00 inch ⁴	16.00 ft
modulus of elasticity	Total load [ASCE7-05 2.3.2]	rectangle - I (sectprop.py)	beam span

